

CLO Trading and Collateral Manager Bank Affiliation

Stavros Peristiani and João A.C. Santos

First draft: January 2017

Abstract

This paper investigates whether the type of institutional affiliation of a collateralized loan obligation (CLO) manager influences the manager's access to information and risk appetite. We base our investigation on CLO managers' trading of distressed loans. Our findings reveal that CLO managers affiliated with banks start to sell off their positions in loans arranged by their bank well before the onset of default and continue to do so even after default. In contrast, CLO managers affiliated with nonbanks do not lower their credit exposures to distressed loans. These findings are consistent with bank-affiliated CLO managers having access to private information but also being more concerned to mitigate the reputational risk exposure of their banking organization. On close inspection, we find that reputational concerns are a significant factor in the decision to divest defaulted loans. Although the value of private information after default is moot, we find that bank-affiliated CLO managers are significantly more averse to holding their own arranged problem loans than holding distressed loans arranged by unaffiliated banks. Besides helping us understand CLO managers' trading activities, our findings also highlight a potential limit to banks' ability to originate loans and distribute them via their affiliated CLOs.

Keywords: CLOs, trading, distressed loans

JEL codes: G21, G23 and G33

Stavros Peristiani: Federal Reserve Bank of New York, 33 Liberty Street, Main 3, New York, NY, 10045 (e-mail: steve.peristiani@ny.frb.org).

João Santos: Federal Reserve Bank of New York & Nova School of Business and Economics, 33 Liberty Street, Main 3, New York, NY, 10045 (e-mail: joao.santos@ny.frb.org).

The views expressed in this paper are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of New York or the Federal Reserve System.

INTRODUCTION

Spurred by several important financial, asset securitization emerged in the past few decades as an alternative method of financing to traditional banking intermediation. Although asset-backed securities were mostly issued in the shadow banking domain, banks and other traditional financial services firms were a dominant force behind the emergence and rapid growth of structured finance (Cetorelli and Peristiani 2012).

Securitization has considerably changed the intermediation of corporate credit with the advent of financial arrangements such as collateralized debt obligations (CDOs) or collateralized loan obligations (CLOs). Over time, banks increasingly used these asset-backed structures to advance their “originate to distribute” approach to lending. The volume of securitization came to a halt during the financial crisis with the collapse of the subprime MBS sector. In contrast with non-agency MBS, whose issuance volume buckled under the weight of the subprime failures, CLO issuance has gradually recovered and remains a significant source of corporate credit finance.¹

Despite the considerable volume of corporate loans outstanding currently in CLOs, we still know very little about the management of these asset-backed assets.² This paper aims to reduce this gap in the literature by investigating how managers of CLOs administer their loan portfolios. CLO managers typically have a “parent” financial institution with which they are affiliated. We are particularly interested in finding out whether CLO managers affiliated with banks manage their loan portfolios differently from CLO managers affiliated with nonbank financial institutions.

¹ The recent rebound of CLO issuance is discussed more extensively in Peristiani and Santos (2015).

² As of the second quarter of 2016, the outstanding volume of CDOs issued in the United States was about \$650 billion (SIFMA).

The affiliation with a bank that acts as a lead arranger in the syndicated loan market may play a role in the way the CLO portfolio is run. All syndicate participants, including CLOs receive borrower information on a regular basis describing their financial performance and covenant compliance. However, loan arrangers, tasked with monitoring the borrower, compile useful private financial information throughout the duration of their relationship with the borrowing firm (Petersen 2004). Therefore, CLO managers affiliated with banks may have access to private information with regard to their investments in those loans that were arranged by their affiliated bank.

While CLO managers can benefit from the informational advantages of bank affiliation, they could also be restrained by the reputational concerns of their parent organization. Bank-affiliated CLOs might be compelled to be more risk-sensitive for the simple reason that their affiliates are more conservative financial institutions subject to prudential regulation and supervision. Implicit in this argument is the assumption that this association could expose the parent organization to reputational risks because its brand is often used to market the CLO securities to institutional clients (Brown 2007). The parent organization could face several perils from badly performing CLOs: litigation risks, regulatory risks, negative investor reaction, and loss of business from clients. Indeed, during the 2007–09 financial crisis, banks’ decision to protect institutional investors and clients from losses stemming from their sponsored asset-backed vehicles underscores the perceived consequences of these reputational costs.

Building on these potential competing effects of CLOs’ affiliations with banks, we investigate two questions in this paper. First, do bank-affiliated CLOs have an informational advantage on loans arranged by their affiliate bank? Second, do bank-affiliated CLO managers have a lower risk appetite?

To investigate our two hypotheses, we estimate a panel regression model that examines how bank-affiliated CLO managers trade loans around the time of their default. We focus on the time before and after default because private information is more valuable during turbulent periods leading up to the default. Further, the focus on default allows us to better discern any concerns that the CLO might engender reputational risks to their sponsor or parent organization stemming from the underperformance of the portfolio collateral.

To formally investigate the link between private information and CLO trading strategies, we start by identifying CLO managers affiliated with banks that are lead arrangers in the syndicated loan market. We also consider other potential tiers of private information stemming from syndicate participations or the CLO underwriter. For example, a possible source of information for collateral managers is the CLO underwriter that is enlisted to help with the selection of collateral, the design of tranches, marketing, and placement with investors. Since CLO underwriters can also serve as syndicate lead arrangers, they can potentially act as an alternative channel of private information to CLO managers. The flow of information from an underwriter, however, is unlikely to be as strong as that from the CLO parent bank because the two entities are not part of the same organization.

This matching process allows us to classify all of the CLO managers' loan positions into three types of relationship to the manager: (1) *lead-arranger relationship*: loans arranged by the bank that is affiliated with the CLO manager, (2) *underwriter/participant relationship*: loans arranged by the CLO underwriter or loans with participation by either the CLO manager or underwriter, and (3) *non-relationship*: loans neither arranged by nor participated in by the CLO manager or underwriter.

Our regression results reveal no significant decrease in CLO managers' holdings of distressed non-relationship loans and only minor changes in CLOs' holdings with indirect

relationships with the arranger of distressed loans around the default date. In contrast, CLO managers affiliated with the loan-arranger of the distressed loan start to sell off their positions well before the onset of default. Interestingly, these managers continue to divest their exposures to distressed loans after default. We find similar results when we investigate CLOs' trading around rating downgrades (that do not trigger a default), affirming bank-managed CLOs' sensitivity to a deteriorating credit risk outlook.

These regression results are quite robust. They continue to hold when we restrict our analysis to defaulting loans held only in the portfolios of CLOs affiliated with loan arrangers, an approach that mitigates sample selection bias. They also remain unchanged when we limit the sample to single-loan positions to reduce the potential mismatch of aggregating multiple loan securities for a given borrower. Further, our empirical findings are robust to different time windows around the event of default.

Our evidence that CLO managers affiliated with the lead-arranger of the distressed loan start to sell off their positions well before the onset of default is consistent with the idea that this affiliation provides access to insightful information. However, our finding that these CLOs continue to sell off their distressed positions after default when the news of insolvency was by then public suggests that concerns with the reputation of the parent bank, rather than access to private information, may explain the trading decisions of these bank affiliated CLOs. The regulated organizational structure of banks and the looming clientele concerns may compel these managers to be more proactive in selling and/or marking down losing portfolio positions in order to protect the reputational value of the organization. Since these risk aversion incentives may also spur better informed managers to start divesting their problem positions before default, we cannot preclude the possibility that these bank-affiliated managers are simply using their superior information to avoid pecuniary losses.

We attempt to disentangle the private information and reputational risk hypotheses by comparing the pre-default trading of bank-affiliated and nonbank-affiliated CLO managers on borrowers that they have no loan relationships with, thus removing private-information biases. In this case, bank and nonbank affiliated CLO managers essentially must depend on available public sources and syndicated loan reports provided by an unaffiliated lead arranger to trade. Although they use similar information, we find that bank-affiliated CLO managers are significantly more responsive to default selling off their distressed loans. This outcome suggests that reputational concerns, rather than access to private information, explain the more conservative reaction of bank-affiliate CLOs in response to increases in risk.

As an additional test, we compare CLOs' post-default trading on loans arranged by the CLO affiliated bank with their trading on defaulted loans arranged by other banks. This comparison reveals that CLOs are more proactive in divesting their bank's defaulted loans than other unaffiliated defaulted positions. Since this trading difference occurs after default, it does not stem from any informational advantages that CLO managers may have vis-à-vis their investments. Instead, this difference in divestments adds further support to the idea that CLOs affiliated with banks, face a greater reputational stigma and are therefore are more conservative.

In sum, our findings reveal that CLOs affiliated with banks are more prudent in trading loans. They are more eager to divest defaulted loans, especially those that were arranged by their affiliated banks. Our evidence that bank-affiliated CLOs start to sell off distressed loans arranged by their parent bank before the onset of default, while consistent with an informational advantage it could also be the result of their lower risk appetite.

Our paper adds to the limited existing literature on the management of CLOs. Despite the critical role that CLOs play in corporate loan securitizations, researchers to date have paid

limited attention to the management of these financial vehicles, instead focusing on other issues.³ The paper in the literature closest to ours is probably Lou, Loumiotis, and Vasvari (2014), which examines the influence of private information on CLO performance. Their findings suggest that CLO managers produce and disseminate private information that they utilize to trade strategically to boost performance.

We, too, are interested in the link between private information and CLO managers' trading strategies, but we also consider the influence of reputational effects derived from the managers' bank affiliations. More important, however, is our more direct way of isolating CLO managers with access to private information. Lou, Loumiotis, and Vasvari (2014) use a regression model to separate public from private information. That approach is inevitably hampered by misspecification errors, as it is very difficult to control for all public sources of information. In contrast, we rely on the nature of CLO managers' relationships with loan arrangers to determine whether the managers have access to private information.

The remainder of the paper is organized as follows. Section 2 details the motivation for our analysis and lays out our hypotheses. Section 3 presents our data and methodology and describes the CLO sample. Section 4 reports and discusses our findings on the impact of CLO managers' affiliation with banks on their trading activities. Section 5 concludes with some final remarks.

1. MOTIVATION AND HYPOTHESES

The key player in a CLO structure is the *collateral manager* (or CLO manager), who is tasked with assembling the CLO portfolio and managing collateral risks throughout the life of the CLO. The collateral of a typical cash-flow CLO usually consists mainly of business loans (syndicated

³ These include the rapid growth of CLOs (Bord and Santos 2012; Guo and Zhang 2015), the impact of securitization on the cost of bank credit (Nadauld and Weisbach 2014 ; and Bord and Santos 2015), the performance of securitized corporate loans (Shivdasani and Wang 2011; Benmelech, Dlugosz and Ivashina 2012 ; Bord and Santos 2015), and the ability of borrowers to renegotiate their loans (Paligorova and Santos 2016).

term loans); a small fraction of it may be corporate bonds and, less commonly, other asset-backed securities. CLO managers receive a fixed management fee plus a contingent fee paid only after the equity tranche achieves a certain hurdle rate.

During the buildup phase, the CLO manager enlists an underwriter, who takes on the traditional responsibilities of assessing investor interest, selling the securities, and providing liquidity support. The underwriter may also facilitate the buildup of the portfolio by offering the manager the option of purchasing loans from a warehouse facility, a purchase often financed with a bridge credit arrangement. The underwriter, in turn, engages a rating agency to determine the seniority tranche structure of the deal and issues the securities (CLO notes and an equity tranche) to investors. The seniority guidelines of the CLO specify how proceeds from principal and interest payments are distributed.

After the ramp-up period, the CLO enters the reinvestment phase, which is followed in turn by amortization. During the reinvestment phase, which ranges from 3 to 5 years, CLO managers reinvest interest proceeds and principal repayments; they also trade assets to (1) realize capital gains, (2) take advantage of good investment opportunities, or (3) avoid potential credit risks. During the following amortization period, the cash flows are used mainly to pay down the security notes. Besides receiving a fixed management fee, CLO managers receive a contingent fee paid only after the equity tranche achieves a certain hurdle rate.⁴ This fee incentivizes managers to deliver a minimum level of returns to the investors in the equity tranche.

CLO managers are bound by number of operational constraints on their trading. In general, they can trade only up to 20 percent of the portfolio's par value each year. They must also meet certain compliance tests (for instance, an overcollateralization asset-liability threshold) and

⁴ The manager receives a portion of its fixed fee before the most senior tranche of debt receives a return or is paid, and it receives the remainder of the fee is after the debt tranches are paid but before the equity tranche is paid.

covenants established to protect investors. However, CLO managers appear to have developed ways to circumvent some of these rules. For instance, Loumioni and Vasvari (2016) note that, to avoid violating overcollateralization requirements, CLO managers often inflate the fair market value of loan assets and prefer to rebalance their portfolio by selling high-quality assets.

CLO managers are usually affiliated with a bank or with a nonbank financial institution such as a hedge fund, private equity firm, or insurance firm. We posit that the type of institution with which the manager is affiliated is potentially important for two reasons. First, it may affect the information available to the manager and by consequence its trading activities. Second, the nature of the parent institution, in particular its ability to take on risk, may influence the risk appetite of the manager.

2.1 The Effect of Affiliation on Availability of Information

Arrangers are responsible for collecting detailed financial information on borrowing firms and for monitoring their compliance with covenants, including their financial performance obligations and they are expected to regularly share that information with syndicate members. It is believed, however, that arrangers also have access to additional information on borrowers that is not shared with other syndicate members.

As noted by Petersen (2004), arrangers gather a good deal of private qualitative information—personal observations, evaluations, viewpoints, and opinions compiled by the borrowers' loan officers over the length of the lender-borrower relationship. This information is important to the understanding of large, complex entities as well as to the evaluation of smaller, less transparent firms. But as “soft,” impressionistic information, it is difficult to transfer to other parties and therefore has the potential to create an informational wedge between arrangers and syndicate participants.

The relationship between borrowers and lenders is bound by confidentiality agreements. For example, the monthly financial loan disclosures, projections, and other financial information certifying the borrower is compliant must remain out of the public domain.⁵ However, the rapid growth of the secondary loan market over the past decade has attracted nonbank and unregulated investors and blurred the limits on the disclosure of private information.

This changing landscape of the syndicated loan market has significantly altered the diffusion of information among market participants. A 2011 report by Standard and Poor's highlighting a rise in loan price volatility in the secondary market supports anecdotal evidence that nonbank syndicate participants have traded on private information. A number of academic studies also find empirical support for the anecdotal evidence. For example, Bushman, Smith, and Wittenberg-Moerman (2011) find that price discovery is faster in loans with covenant violations (which are disclosed to syndicate members) and find a stock price reaction to these private disclosures. Massoud, Nandy, Saunders, and Song (2009) argue that investors benefit from using insider information obtained from participating in syndicated loan activities and renegotiations.

CLO managers purchasing a loan are entitled to the customary financial reports available to all members of the syndicate. Because syndicate arrangers have proprietary borrower information, it follows that CLO managers affiliated with the loan arranger may also be privy to this unique information and use it to enhance their trading activities. This gives us our first testable conjecture:

HYPOTHESIS 1: CLO managers affiliated with loan arrangers have access to private information.

2.2 The Effect of Affiliation on Risk Appetite

⁵ In the case of bonds issued by public firms, information about the bonds is eventually disclosed in reports to regulators.

A CLO manager's risk appetite will likely be influenced by concerns for the reputation of its parent. Brown (2007) argues that, in today's information-driven financial environment, reputational challenges are the most significant management risks faced by banking firms. When the CLO manager is affiliated with a bank, the manager's concerns for its parent's reputation can be amplified because banks are generally more averse to reputational risk than other financial institutions.⁶

A large literature documents the importance of reputation for banks. For example, Carter and Manaster (1990) and Carter, Dark, and Singh (1996) show that initial public offerings (IPOs) of equities underwritten by top-tier (that is, more reputable) banks exhibit stronger long-run returns, suggesting these banks have a competitive edge in the equity underwriting market. Reputation also influences the effectiveness of banks when they act as loan arrangers. Gopalan, Nanda, and Yerramilli (2011) report that loan arrangers whose syndications history indicates a relatively high number of bankruptcies after syndication are forced to retain a bigger share of the loan; the study also finds that arrangers with a history of poor performance are less able to attract new syndicated loan borrowers and participant lenders.

In the securitization framework, concerns with the reputation of the parent bank will likely shape trading strategies of affiliated CLO managers because a poorly performing asset-backed security has the potential to undermine the reputation of the bank parent. The recent decision of many large banks to close down their troubled subprime mortgage lending subsidiaries and securitization desks is a good example of institutions' concerns with reputation risk. Similarly, after the collapse of the market for equities of Internet-related firms in 2000, many banks and

⁶ Broadly speaking, an organization's reputational risk refers to problems stemming from negative public opinion about its operations. For a bank, such negative opinion can harm its ability to maintain and expand its client base. Incidents that damage its reputation can also expose the bank to costly litigation and increased regulatory scrutiny that reduce profitability and diminish franchise value.

investment banks eliminated their underwriting desks and brokerage units that were tarnished by the market meltdown. These considerations bring us to our second hypothesis.

HYPOTHESIS 2: Reputational concerns reduce the risk appetite of CLO managers affiliated with a bank.

3. METHODOLOGY, DATA, AND SAMPLE CHARACTERIZATION

3.1 Methodology

The first step in our methodology is to specify a measure of CLO trading activity. To that end, let P_{tij} denote CLO (j) holdings in loan (i) at month (t), with the monthly time indicator ranging from $t=0$, the origination month, until T . A common measure of trading activity for loan (i) is given by the change in holdings, or ΔP_{tij} , representing month-to-month changes.⁷ A better estimate of the cumulative trading activity is given by $R_{tij} = P_{tij}/P_{0ij}$, where P_{0ij} represents the initial investment. We formally refer to R_{tij} as the *investment ratio* of CLO (j) in loan (i) at month (t).⁸ We set the investment ratio at 100 percent at ($t=0$). Our analysis excludes the ramp-up phase of trading so as to avoid the unusual trading volatility of that period and its impact on R_{tij} .

To analyze the importance of private information, we use a straightforward regression specification to analyze CLO managers' trading around the event of borrower default. We assume that a CLO investment ratio in a borrower R_{tij} is a function of the borrower characteristics, the structure of the CLO, and a set of CLO-manager-specific factors. More

⁷ CLO managers also invest a small fraction of their portfolio in corporate bonds and other asset-backed securities. Our analysis, however, focuses only on the more important loan assets.

⁸ We also explored alternative measures of trading activity, such as the simple change ΔP_{tij} and $R_{tij} = \sum_t \Delta P_{tij}/P_{0ij}$. Overall, the regression analysis results were very similar.

precisely, the model is defined

$$\text{by } R_{ij} = \alpha_t + \alpha_i + \beta_1 x_{ij}^B + \beta_2 x_{ij}^{CLO} + \sum_{h=-H}^H \delta_h^M \text{Rel}_{ij,h}^M + \sum_{h=-H}^H \delta_h^U \text{Rel}_{ij,h}^U + \sum_{h=-H}^H \delta_h^N \text{NOrel}_{ij,h} + \varepsilon_{ij}. \quad (1)$$

The explanatory variable, α_t , is a binary indicator of (yearly) time variation; α_i controls for borrower fixed effects; and ε_{ij} represents the random error. The vector x_{ij}^B includes publicly available borrower-specific measures that vary over time (for instance, a borrower's monthly credit rating); similarly, the explanatory vector x_{ij}^{CLO} represents CLO-specific explanatory variables, such as the size of the CLO facility.

We distinguish the CLO manager's access to borrower information using the binary variable $\text{Rel}_{ij,h}$, which indicates the nature of the relationship the CLO manager has with the arranger of the defaulting loan(h) months from the time of default/distress.⁹ For simplicity, we assume that $h = -H, \dots, 0, \dots, H$, where ($h = 0$) represents the month of the default event.

The explanatory variable $\text{NOrel}_{ij,h}$ indicates that the CLO manager (and its bank underwriter) does not have a relationship with the arranger of the defaulting loan. In this case, the CLO manager only receives the customary reports from another *unaffiliated* arranger. Credit agreements mandate that syndicate members be informed of the borrower's financial condition; as noted previously, investors receive monthly reports detailing firm profitability, capitalization, leverage, and net worth. This private information is usually hard in nature and continues throughout the duration of the loan security.¹⁰ Additionally, CLO managers can exploit public

⁹ More precisely, we assert that $\text{Rel}_{ij,h} = 1$ if there was a particular loan-information relationship between the manager and borrower over the past 7 years. The 7-year relationship duration is consistent with the typical maturity of a term-loan facility. Throughout the life of the loan, the loan arranger is responsible for monitoring the financial viability of the borrower and for regularly sharing this information with syndicate members.

¹⁰ Collateral managers typically purchase a loan from a warehouse facility or from the secondary market. Once the loan is added in the portfolio, the collateral manager (formally the CLO) becomes a member of the syndicate and is qualified to receive monthly reports and updates detailing the borrower's financial condition and capacity to meet loan covenant conditions.

information, including credit rating changes, equity analysts, regulatory disclosures, and business-financial intelligence reports, to trade on distressed firms.

The model specification (1) considers two possible sources of *affiliated* proprietary information, represented by $Rel_{tji,h}^M$ and $Rel_{tji,h}^U$. The variable $Rel_{tji,h}^M$ indicates whether the CLO manager is affiliated with the arranger of the defaulting loan; $Rel_{tji,h}^U$ indicates whether the CLO underwriter was the arranger of (or participated in) the defaulting loan. The first of these two variables, $Rel_{tji,h}^M$, captures the instances in which the CLO manager can obtain the most insightful and valuable soft information about the borrower. The second, $Rel_{tji,h}^U$, represents the intermediate case in which the CLO manager could get access to loan information through less direct channels such as through its underwriting bank.¹¹ Arguably, firewall privacy rules may make it difficult for the CLO underwriter to share private information with a CLO manager beyond what is typically furnished to other syndicate members.

These variables, together with the parameter estimates of δ_h^i , tell us the incremental change in the investment ratio (h) months before the event of default by the corresponding information relationship group (i). For example, a value of 5 for δ_h^i corresponds to a 5 percent increase from the initial (t = 0) position; the actual change from, say, a 140 percent investment ratio at month (h – 1) is 5/140, or 3.57 percent.

¹¹ There is a subtle difference between $NORel_{tji,h}$ and $Rel_{tji,h}^U$, the latter representing the case where the source of the information is an affiliated participant of the manager. As noted, the information $NORel_{tji,h}$ stems from customary reports available to all syndicate members, including those managers with a $Rel_{tji,h}^U$ relationship; however, in the latter case, managers could gain additional insights because their participant affiliate or arranger/participant CLO underwriter has extensive syndication experience with the borrower.

We consider two statistical tests for Hypothesis 1 (which, to recall, asserts that CLO managers with loan arranger affiliations gain access to more information, with the implication that they are therefore better able to respond to distress before default):

HYPOTHESIS 1A: CLO managers affiliated with the arranger of the defaulting loan are able to divest from that loan before default. This premise can be statistically tested by maintained

hypothesis $H_0 : \bar{\delta}_B^{\iota} \leq 0$, where $\bar{\delta}_B^{\iota} = [\sum_{h=-1}^{-H} \delta_h^{\iota} / H]$ such that $\iota = M, U$.

HYPOTHESIS 1B: The sell-off before default of distressed loans by CLO managers affiliated with the arranger of that loan is greater than the sell-off by remaining CLO managers. We test this premise by $H_0 : \bar{\delta}_B^M \leq \bar{\delta}_B^U$ or $H_0 : \bar{\delta}_B^M \leq \bar{\delta}_B^N$.

As we noted above, investment decisions in problem loans of CLO managers affiliated with banks may be influenced by concerns related to the reputation of the parent bank. One important artifact of our sample is that the CLO manager affiliation with loan arrangers is dominated by banks. This was to be expected because banks are the dominant arrangers in the syndicated loan market. While CLO managers affiliated with bank arrangers likely have access to more information, they also have to cope with the consequences of their affiliated entity (bank) facing stricter regulatory oversight and market discipline. This makes the task of analyzing the value of bank affiliations more challenging because private information and reputation are complementary in the sense that both would lead to a decrease in the investment ratio before default.

A simple way to disentangle the effect of private information and reputational risk is to examine trading after default, because by then the news is public. The negative news of default are revealed and fully priced by market participants, but conceivably, under the reputational risk

hypothesis, bank-affiliated CLO managers continue to divest these defaulted loans to avoid these costs.

To statistically test Hypothesis 2, we investigate the following two implications of that hypothesis.

HYPOTHESIS 2A: Reputational concerns induce CLO managers affiliated with loan arrangers to sell off defaulted loans after default even though the bad news is fully priced. This premise can be statistically tested by $H_0 : \bar{\delta}_A^M \leq 0$, where $\bar{\delta}_A^M = [\sum_{h=1}^H \delta_h^M / H]$.

HYPOTHESIS 2B: Reputational concerns induce CLO managers affiliated with loan arrangers to divest more of their credit exposures after default when compared with other CLO managers. We statistically test this premise by $H_0 : \bar{\delta}_A^M \leq \bar{\delta}_A^U$ or $H_0 : \bar{\delta}_A^M \leq \bar{\delta}_A^N$.

3.2 Data Sources

To investigate our hypotheses regarding the impact of CLO managers' affiliations on their trading decisions, we start by gathering information from Moody's CDO Services database on the composition of CLO investment portfolios over time.

The Moody's database reports portfolio positions compiled from monthly trustee reports. Because this database does not contain unique loan security identifiers, we aggregated all CLO portfolio security positions at the borrower level. This does not significantly alter the structure of the portfolios because nearly 80 percent of the CLO positions have only one investment per borrower. This aggregation gives us the opportunity to trace in a consistent way the changes in the CLOs' total notional value invested in each borrower over time.¹²

¹² Collateral managers often swap loan securities from the same borrower. Sometimes the replacement results from an exchange, redemption, or restructuring and is thus nondiscretionary. In other instances, managers may prefer a security from the same borrower with better features, such as longer duration and higher seniority. None of these security replacements contribute to turnover in our borrower-specific information.

The Moody's database also provides extensive information on CLO managers and underwriters. This information is critical for our goal of understanding the impact of the relationships between the CLO manager and other players in the securitization chain. We complement these data with information from Capital IQ and Dealscan to ascertain the nature of the relationship between the CLO manager and the members of the syndicate in each loan in the CLO portfolio of collateral.

3.3 Sample Characterization

Our sample consists of 239 CLOs originated between 2007 and 2011 as arbitrage cash flow or small- to medium-market CLOs.¹³ We exclude synthetic transactions that derive their cash flow from credit default swaps and are therefore not actively managed. The average original face value of the sample CLOs is \$485 million (Table 1). They are invested in more than 5,000 borrowers with about 65,000 unique CLO-borrower positions. The overall number of observations in the CLO-borrower panel is about 1.47 million.

CLO managers are affiliated with five broad categories of financial institutions: banks, insurance firms, hedge funds, private equity firms, and investment asset-manager firms. Table 1 reveals a significant heterogeneity among CLO managers. Importantly for our purposes, 55 of the 239 CLO managers are affiliated with banks; equally important, a default occurs during the sample period (2007–11) in more than 20 percent of the borrowers in which the CLOs invest.¹⁴

¹³ Small- to medium-market CLOs are a minor specialized segment of structured corporate debt products typically backed by loans to small and medium-size enterprises.

¹⁴ We use information from several sources to ascertain corporate default. The trustee reports compiled by Moody's identify borrowers that defaulted. In addition, Moody's separately provides its own assessment of whether a borrower has defaulted. We complemented these main sources of default information by identifying (1) distressed borrowers assigned a D rating by any credit rating agency and (2) firms that filed for bankruptcy (Capital IQ). As a final check, we manually checked the entire sample of borrowers using Capital IQ's corporate timeline to verify the sequence of insolvency events. The onset of firm distress is defined as the first occurrence of insolvency reported in the trustee reports, the Moody's database, and bankruptcy information.

Figure 1 shows CLO managers' trading activity over the first four years of the CLO's life. During the ramp-up period (the first six months), the investment ratio varies significantly for all types of manager affiliation. Recall that, by definition, the investment ratio is 100 percent at $t=0$. The greater variance during the ramp-up period is not unusual because managers are building up their portfolios from scratch from the available warehouse facilities and secondary loan market. Nonetheless, to exclude the unusual trading volatility of the ramp-up period and its impact on R_{it} , we omit the ramp-up phase of trading in our regression analysis.

CLO managers continue to trade actively well beyond the ramp-up period (Figure 1), with the average investment ratio slowly rising from around 110 percent at $t=18$ to around 140 percent at $t=36$. The trading pattern naturally varies over the CLO lifecycle.¹⁵ For a sample of arbitrage cash-flow CLOs similar to the present sample, Peristiani and Santos (2015) estimate the average monthly buying turnover (relative to assets) over the 2007–11 period to be about 5.5 percent (corresponding to a 66.5 percent annual rate). After the ramp-up period, managers become more active sellers as they seek to replace some of the original temporary holdings. Although part of the position turnover stems from loan repayments and renegotiations, most of the buying and selling is discretionary.

Table 2 decomposes the sample of CLO-borrower holdings for the subset of firms that defaulted, which forms the basis of our regression sample. Close to 69 percent of the CLO borrowers have no affiliation with either the manager or underwriter because typically the portfolios are compiled from various loan warehouse facilities; moreover, a lot of the CLO collateral consists of smaller low-rated firms with a limited history of syndicated loan issuance.

¹⁵ In theory, a CLO with fully invested proceeds must sell securities in its portfolios to reinvest. Typically, before CLOs enter the reinvestment period, CLO buying precedes selling as collateral managers accumulate positions.

The remaining segment of borrowers, however, has an information relationship with manager or underwriting allowing us to empirically test our maintained hypotheses.

Table 3, in turn, summarizes the control variables of our regression analysis, which models a CLO manager's decision to modify its holdings of loan asset (i) at time (t) represented by investment ratio R_{ijt} . As noted earlier, a small portion of the collateral in a typical cash-flow CLO consists of corporate bonds and, occasionally other structured securities. Given the focus on evaluating the importance of private syndicate information, we restrict the analysis to loan positions, which constitute the bulk of the collateral. The average investment ratio for the regression sample is about 137 percent. In some extreme cases, in which the initial holdings are very small, the investment ratio can be much larger. We mitigate the impact of these outliers by winsorizing the investment ratio, as well as other explanatory variables in the regression, at the 1st and 99th percentiles.

The explanatory variable $\log(\text{POSITION})$ controls for the size effect of loan holdings. The average position in each borrower is around \$3.1 million, with the 99th percentile at more than \$19 million. The relatively small nominal position and narrow range in positions show that CLOs are well diversified, holding investments in about 160 unique borrowers. The variable $\log(\text{MATURITY})$ also controls for the remaining maturity of the loan. The median volume-weighted maturity in our sample of loans is 4 years. Granted that the typical collateral maturity for most CLO tranches is around 9 years, managers will sometimes find it more difficult to reinvest in newly issued syndicated deals that may create a duration mismatch. About 95 percent of the firms in the portfolios are domiciled in United States (US_FIRM).

The borrower's credit rating is a key control in the regression model because it measures the impact of public information on the investment ratio. The average borrower credit rating of B+

actually masks the true credit quality risk, as more than 55 percent of the borrowers fall in the range of B+ to B-. In light of the discrete nature of ratings, the regression model includes a set of dummy controls, one for each rating. Under the maintained hypothesis that the investment ratio is risk sensitive, the coefficients of these binary credit indicators are expected to transition from positive to negative as managers dispose of riskier securities before default.

The bottom panel of Table 3 summarizes the relationships between the borrowing firm, the CLO manager, and its underwriter. The average for $\text{NORel}_{ji,h}$ is 68 percent, indicating that many CLO managers have only “weak” relationships with the borrowers they invest in. Most of these relationships derive from the CLO manager being a syndicate participant, or from its underwriter being a participant in the loan the CLO invests in, or both. On average $\text{Rel}_{ji,h}^U$ is equal to 29 percent, with 9 percent obtained from syndicate participants.

CLO managers have the most informative link with borrowers when the manager is the arranger of the loan, which occurs on average for only 3 percent of the borrowers. The small fraction $\text{Rel}_{ji,h}^M$ is not surprising considering that banks are the dominant arrangers in the syndicated loan market and thus only CLOs affiliated with banks can have these relationships.

4. DETERMINANTS OF CLO TRADING

The regression model defined by equation (1) outlines several determining factors for the trading activities of CLO managers; the primary focus, however, is on their affiliation with arrangers of loans before and after their default. One way to estimate equation (1) is to use the entire panel encompassing all CLO $\times$ borrower observations for all possible monthly observations before and after default. This approach is not very practical because the monthly panel of distressed borrowers is distributed asymmetrically around default. For example, in about 10 percent of the borrower defaults, CLOs have no previous exposure to the borrower; rather, in

those cases, the CLO manager purchased the loan after the borrower defaulted. These uneven investment patterns before and after default introduce sample-selection biases because the investment strategies of managers before default would differ from those after default. To better balance these investment incentives, we start by focusing on defaults for which CLOs had a minimum of 6-months' investment history before and after the default date.

4.1 Testing Hypothesis 1: The Importance of Private-Information Relationships

We begin with a graphical overview of CLO managers' responses to borrower distress, that is, the in-sample regression investment-ratio predictions for the loan-relationships alternatives (Figure 2). This graphical evidence tracing the response to private information offers a useful summary of CLO managers' trading activities around default.

The investment ratio forecast for the relationship between the loan arranger and the CLO manager is calculated for the average borrower in the sample, defined more formally as $\hat{R}_h^i = \bar{\alpha}_i + \bar{\alpha}_j + \bar{\alpha}_t + \hat{\beta}_1 \bar{x}^B + \hat{\beta}_2 \bar{x}^{CLO} + \hat{\delta}_h^i = \hat{\mu} + \hat{\delta}_h^i$, such that $i = N, M, U$ and $h = -12, \dots, 12$. The estimates $\bar{\alpha}_i$ and $\bar{\alpha}_t$ represent the average regression for borrower fixed-effects and time effects, respectively. The first five components of the investment ratio forecast represent the average-response baseline $\hat{\mu}$; the extant impact of the private-information relationships is determined by the corresponding $\hat{\delta}_h^i$ deviation coefficients.

When CLO managers have only a weak relationship with defaulting borrowers ($Rel_{ij,h}^U = 1$), either via the CLO underwriter or through their participation in the loan syndicate, we observe only about a 2 percent decline in distressed loans before default (Figure 2), and no significant trading occurs when CLO managers have no relationships with defaulting borrowers ($NORel_{ij} = 1$). But when CLO managers are affiliated with the arranger of the defaulting loan

($\text{Rel}_{ij,h}^M = 1$), they gradually lower their investment ratio to distressed firms starting about one year before default. By the time of default, these managers have reduced their exposure by 14 percent. Interestingly, they continue to divest thereafter, decreasing their overall holdings from month $h = -12$ to $h = 12$ by 33 percent.

While this disparate pattern in the investment ratio between manager-affiliated and non-affiliated loans is consistent with our premise, an alternative interpretation is that the decline does not stem from a loan sell-off but simply because managers mark these securities to market. CLO collateral normally value securities at par to avoid the unnecessary problems that come with mark-to-market volatility. However, CLO managers are expected to re-price their collateral in the event of default at the fair market value or recovery value. Arguably, the apparent drop in investment ratio may stem from loan re-pricing after default; however, this explanation is paradoxical, as our graphical evidence would suggest that mark-to-market rules are not consistently applied.

In addition to providing information of CLO positions, the Moody's database also includes a separate module tracing the CLO's trading transactions. One drawback with the transactions data is that it is mostly compiled from independent trustee reports without common loan identifiers to link with the positions information. Nevertheless, using the available loan identifier or loan facility names, we were able to merge a significant segment of the CLO positions and transactions data. Overall, this exercise reveals that most of the nontrivial changes in loan positions are associated with specific trading transactions and are therefore not caused by market value accounting. It is also interesting to note that managers sometimes report the primary reason for the transaction. About 82 percent of the trading activity is discretionary and is related to the credit deterioration or improvement of the loan asset. The remaining trading activity is

nondiscretionary, triggered mostly by corporate restructuring, debt refinancing, redemptions, and loan exchanges.

The graphical evidence of the trading responses of CLO managers does not, however, assess the statistical significance of the changes in the investment ratio. To that end, we present in Table 4 the regression results determining the effect of CLOs' affiliations on the investment ratio for a 12-month horizon before and after default ($H = 12$). We first focus on the bottom part of the table that formally investigates the statistical significance of the private-information hypotheses outlined in the previous section. The table presents the relevant change for the specified hypothesis and indicates the statistical significance of the underlying F-test.

The first test considers whether different types of relationships influence CLO managers' trading activity before default (Hypothesis 1A). The maintained hypothesis is that CLO managers would not respond to private news before default, that is $H_0 : \bar{\delta}_B^1 = 0$.¹⁶ As suggested by the graphical analysis (Figure 2), the change in the pre-default investment ratio for CLO managers that have a relationship with borrowers only via their underwriter or syndicate participation ($Rel_{ij,h}^U$) and for those with no relationships ($NORel_{ij}$) are not statistically significant ($\bar{\delta}_B^U = -1.56$ and $\bar{\delta}_B^N = 2.62$, respectively). The negative response on $\bar{\delta}_B^U$ suggests that CLO managers may get some soft information through their underwriters or their syndicate participation, but the impact of these relationships is statistically insignificant.

¹⁶ Although we formally described the hypothesis using a one-sided framework, in practice the statistical tests need to consider the two-sided possibility that the investment ratio can increase or decrease during the specified period. The F-test is therefore designed to reject the null hypothesis that the trading response δ_B^1 is zero in favor of the alternative that it is either positive or negative.

In contrast, the average decline in this 12-month period before default for CLO managers affiliated with the arranger of the defaulting loan ($Rel_{ij,i,h}^M$) is $\bar{\delta}_B^M = -11.70$ and statistically significant.

Hypothesis 1B proposes that the sell-off of distressed loans by CLO managers that are affiliated with the loan arranger is greater before default than is the sell-off before default by the remaining CLO managers. Indeed, the average difference between the investment ratio for borrowers with CLO manager-loan arranger relationships and those without any relationships is $\bar{\delta}_B^M - \bar{\delta}_B^N = -14.32$ and statistically significant (Table 4, bottom panel).

The strong statistical decline in the investment ratio in the period leading up to default is consistent with Hypothesis 1, which is that CLO managers affiliated with the arranger of the defaulting loan use their private information to lower their exposure to distressed borrowers before default.

4.2 Testing Hypothesis 2: The Importance of Reputation Risks

We examine the statistical significance of Hypotheses 2A and 2B by comparing the post-default trading response. The F-tests presented at bottom panel of the table formally reveal a large and statistically significant difference between the relationship comparisons. In particular, we observe large negative gaps ($\bar{\delta}_A^M - \bar{\delta}_A^N = -22.50$ and $\bar{\delta}_A^M - \bar{\delta}_A^U = -21.58$) for CLO managers that have relationships with arrangers of the defaulting loans. These findings are consistent with the reputational risk premise because, by that time, most bad news is fully priced by the market.

Regression results show that CLO managers affiliated with loan arrangers lower their exposure to distressed positions before and after default in comparison to non-relationship managers. The results thus indicate that those CLO managers are averse to deteriorating credit conditions. The motive for such divestment, however, is not clear at this stage. Is divestment

driven by access to private information, or by reputational concerns stemming from the fact that the vast majority of loan arrangers are banks, or perhaps both? For sure, private information should not be a significant factor after default because negative news is fully priced. While the divestiture after default among CLO managers affiliated with loan arrangers is consistent with Hypothesis 2, we cannot dismiss the value of private information allowing managers to trade losing positions before default.

In Section 4.4, we examine a number of model specifications that attempt to unravel the importance of these competing explanations. But first we discuss the findings of a set of robustness tests we carried out on our initial findings revealing an significant aversion by bank-affiliated managers to risky investments.

To mitigate possible adverse selection loan biases between the relationship groups, we restrict our sample to loans that were granted by the bank arrangers affiliated with the CLOs in our sample (Table 4, second column). . As expected, the sample of borrowers in this panel is notably smaller, declining from 374 firms to 67 firms. However, the regression results for this subset are very similar, suggesting no discernible sample bias.

Another alternative sample specification focuses on the more homogeneous, single-loan borrowers to lessen possible mismatch problems stemming from loan aggregation. Recall that, to avoid the problems stemming from the absence of loan identifiers in our data, we focus on CLO managers' overall investment at the borrower level. Restricting the sample to borrowers that have only one loan assures us that CLOs are investing in the same loan. This restriction does not affect the regression coefficients and related F-tests (Table 4, last column).

A final robustness check estimates the investment ratio response around a narrower 6-month default horizon. In this case (Table 5), the regression results are very similar to those we obtain

with the 12-month window, indicating that much of the divestiture in distressed borrowers occurs closer to the default date.

4.3 Influence of Other Factors on CLO Trading

The top part of the Table 4 presents the impact of the different controls. We briefly discuss here the impact of the remaining explanatory variables that we consider in our investigation of CLOs' trading around loan defaults (Table 4, top panel). The large and statistically significant impact of $\log(\text{POSITION})$ signifies that a 10 percent increase (decrease) in borrower holdings would increase (decrease) the investment ratio by about 8 percent. The coefficient of MATURITY is negative, affirming that collateral managers are more inclined to avoid longer maturity loans. While longer duration assets are generally more attractive to institutional investors, collateral managers' portfolio choices might be constrained by guidelines in the CLO indenture that govern weighted average life thresholds.

The coefficient of the size of the asset-backed portfolio measured by CLO_BALANCE is negative, indicating that larger CLOs have lower investment ratios for defaulting borrowers. Although there are some conceivable interpretations for this negative effect, this outcome may be an artifact of the CLO design. Larger CLOs are intrinsically more diversified, which allows them to maintain lower and perhaps more optimal investment ratios over the life of the security.

As expected, credit ratings strongly influence the manager's willingness to keep a loan. This is evidenced by the estimated parameters on the credit rating dummy variables, which change from positive to negative as the loan credit scores deteriorate (from about 20 percent for loans rated between [AAA-BBB-] to nearly -14 percent for distressed loans with ratings in the [CC-C] bracket).

Finally, our results show that CLO managers make larger investments in the loans of borrowers in which they also have an equity investment. The positive and statistically significant coefficient of EQUITY_REL indicates that CLO managers retain on average about 15 percent higher investment ratio on those borrowers that they also have an equity investment.¹⁷ This investment ratio commitment remains unchanged before and after default.

4.4 Disentangling the Informational and Reputational Hypotheses

The regression results discussed so far reveal that CLO managers with access to informative loan relationships are more proactive in disposing of distressed loans both before and after default. Private information, however, should not be a significant factor after default because by that time all negative company news is public; instead, the post-default divestiture is more consistent with the reputational hypothesis faced by bank-affiliated CLOs. Despite the compelling logic of this interpretation, demonstrating it empirically is difficult. The disposal of defaulted loans by bank-affiliated CLOs may simply represent opportunistic trading unconnected to any reputational concerns. They may also be more inclined to alleviate reputational problems regardless of their access to private information. That is, they could simply be selling underperforming assets to avoid unnecessary scrutiny.

To unravel the intertwined influences of information and reputation, we investigate two versions of our regression model to distinguish the response to distress by bank-affiliated CLO managers from the response of other CLO managers. The first version allows us to compare the reaction of bank- and nonbank-affiliated CLOs before and after default without controlling for the presence of loan relationships:

¹⁷ The equity relationship variable is a binary indicator of firm ownership constructed from Capital IQ company information. The equity ownership measure reflects primarily situations where the manager is a private equity owner or a current stakeholder in the firm but excludes 13F share ownerships of listed companies reported to the SEC.

$$R_{tij} = \alpha_i + \alpha_t + \beta_1 x_{tij}^B + \beta_2 x_{tij}^{CLO} + \sum_{h=-H}^H \delta_h^{BANK} Rel_{tij,h}^{BANK} + \sum_{h=-H}^H \delta_h^{NONBANK} Rel_{tij,h}^{NONBANK} + \varepsilon_{tij}. \quad (2)$$

The underlying premise here is that bank-affiliated CLOs might behave more prudently so as to avoid reputational risks. Using the framework of Hypothesis 1, we can investigate whether bank-affiliated CLO managers proactively divest distressed loans before default by testing $H_0 : \bar{\delta}_B^{BANK} \leq \bar{\delta}_B^{NONBANK}$; similarly, Hypothesis 2 can be adapted to test whether bank-affiliated CLOs facing greater reputational constraints continue disposing of problem loans after default by testing $H_0 : \bar{\delta}_A^{BANK} \leq \bar{\delta}_A^{NONBANK}$.

The estimates of equation (2) reveal a significant difference in the response between and bank- and nonbank-affiliated CLO managers (Table 6, first column). Bank-affiliated CLO managers decrease their average pre-default exposure by 5.11 percent ($\bar{\delta}_B^{BANK} = -5.11$) and continue to divest from these insolvent borrowers after default by 4.74 percent ($\bar{\delta}_A^{BANK} = -4.74$). In contrast, nonbank-affiliated CLO managers slightly increase their post-default investment ratio in distressed borrowers. The statistically significant difference in the post-default trading response highlights the importance of the regulatory structure and corporate governance under which bank- and nonbank-affiliated CLO managers operate.

The results just presented are based on the entire sample of borrowers and thus include borrowers that have relationships with the CLO manager. The presence of those borrowers could bias the regression estimates because it may affect, for example, the information available to the CLO manager. To address this concern, we restrict our sample to the subset of borrowers that *do not* have loan relationships with the CLO manager or its underwriter. For this subsample of borrowers, bank- and nonbank-affiliated CLOs rely on information provided to all syndicate participants by an unaffiliated loan arranger and other public sources.

The regression findings based on this non-relationship subsample reveal that bank-affiliated CLOs are more responsive to default (Table 6, second column). Over the pre-default and post-default periods, bank-affiliated CLOs without any relationships lower their default exposure by 1.92 percent and 2.66 percent, respectively ($\bar{\delta}_B^{\text{BANK}} = -1.92$ and $\bar{\delta}_A^{\text{BANK}} = -2.66$). Although the economic response is economically smaller, the difference between bank- and nonbank-affiliated managers is statistically significant. For CLO managers affiliated with the loan arranger, this finding suggests that the sell-off of distressed loans in the period leading up to the default date, while consistent with their having access to private information, may also be induced by reputational concerns stemming from their affiliation with banks, which tend to have a lower risk appetite.

To further disentangle the private information and reputational motives for CLO managers' trading around loan defaults, we apply our second test, in which we estimate a hybrid of the regression specifications (1) and (2):

$$R_{ij} = \alpha_i + \alpha_t + \beta_1 x_{ij}^B + \beta_2 x_{ij}^{\text{CLO}} + \sum_{h=-H}^H \delta_h^M \text{Rel}_{ij,h}^M + \sum_{h=-H}^H \delta_h^U \text{Rel}_{ij,h}^U + \sum_{h=-H}^H \delta_h^{\text{N-BANK}} \text{NOrel}_{ij,h}^{\text{BANK}} + \sum_{h=-H}^H \delta_h^{\text{N-NONBANK}} \text{NOrel}_{ij,h}^{\text{NONBANK}} + \varepsilon_{ij}. \quad (3)$$

This specification decomposes the dummy explanatory variable NOrel_{ij} (for CLO managers without any loan relationships) into $\text{NOrel}_{ij}^{\text{BANK}}$ (for the managers that are bank-affiliated) and $\text{NOrel}_{ij}^{\text{NONBANK}}$ (for the nonbank-affiliated). The decomposition of NOrel_{ij} is useful because there of the significant overlap among CLOs affiliated with arrangers of defaulting loans ($\text{Rel}_{ij}^M = 1$) and the CLOs affiliated with banks that invest in non-relationship

loans ($\text{NORel}_{ij}^{\text{BANK}} = 1$).¹⁸ This overlap provides a useful comparison baseline for testing the value of private information because the trading decisions on loans with and without relationships are made largely by the same core group of bank-affiliated CLO managers subject to the same reputational constraints.

If loan relationships do not offer any informational advantages before default, then bank-affiliated CLO managers should trade relationship defaulting loans and non-relationship defaulting loans in a similar fashion; more precisely, we should expect to find $H_0 : \bar{\delta}_B^{\text{N-BANK}} = \bar{\delta}_B^{\text{M}}$. If, on the other hand, CLO managers trade their relationship loans more aggressively in the period leading up to the default, that will provide support to the idea that they rely on private information because the impact of reputational concerns should be similar across relationship and non-relationship loans. In fact, we find the latter possibility to be the case (Table 7, bottom half): a 12.07 percent decline for CLOs with loan relationships ($\bar{\delta}_B^{\text{M}} = -12.07$) and only a 2.11 percent decline in positions held by CLOs having no explicit relationships with the borrower ($\bar{\delta}_B^{\text{N-BANK}} = -2.11$).

The post-default trading response by this same set of CLO managers is equally informative. We observe a much higher sell-off of relationship loans ($\bar{\delta}_A^{\text{M}} = -19.96$) than non-relationship loans ($\bar{\delta}_A^{\text{N-BANK}} = -3.10$). Considering that these results are driven by a set of about 10 bank-affiliated CLO managers and that the value of private information is fully priced after default and thus moot, the sell-off gap suggests a more nuanced reputational risk reaction. In particular, reputational risks might be greater when the loan is actually arranged by the CLO manager's

¹⁸ The 37 bank-affiliated CLOs in our sample are managed by 21 unique banks (Table 1). Moreover, banks managing loans with syndicated loan relationships overlap significantly with banks managing loans without such relationships. In particular, bank managers that are arrangers (represented by $\text{Rel}_{ij}^{\text{M}} = 1$) hold about 70 percent of the loans without any relationships (indicated by $\text{NORel}_{ij}^{\text{BANK}} = 1$).

bank parent. Our findings regarding bank-affiliated CLO managers reveal that when loans arranged by the parent bank fall into distress, the managers are quite averse to holding them, whereas they are much less eager to sell defaulted loans not associated with the parent bank. These results raise the prospect that arrangers, and by association their affiliated CLO manager, might face greater reputational stigma when the loans they arrange default and therefore, to avoid market scrutiny, they are less likely to keep these problem assets.

Our findings are consistent with several studies in the banking literature highlighting the importance of reputational and operational risk problems. Gopalan, Nanda, and Yerramilli (2011) show that loan arrangers experiencing greater bankruptcies suffer lower rates of participation by investors and attract fewer syndicated loan borrower clients. Perry and de Fontnouvelle (2015) find a significant negative market reaction to the announcement of internal fraud losses for firms with strong shareholder rights. Large banking firms that crucially value their client base and brand would therefore be more vulnerable to internal reputational mishaps and negative firm perceptions that could engender an adverse response from investors and clients.

4.5 CLO Trading Around Credit Downgrades

The primary focus of our analysis so far has been estimating the trading activity of CLO managers around borrower default. Our findings reveal that bank-affiliated managers are more proactive in lowering their exposure to loans arranged by their parent bank both before and after their default. To consider the response of CLO managers to borrower developments less severe than default, we examine credit rating downgrades as a signal of borrower distress.

The credit rating downgrade sample is very useful in several dimensions. To avoid credit downgrades that are a precursor of firm insolvency, we eliminate from our analysis all defaults occurring in the sample of firms with downgrades. Thus, the regression estimates the trading

activities of CLO managers on firms experiencing rating slipups but not falling into default. In contrast to defaults that often correspond to a terminal event for a struggling firm, credit rating downgrades are a less dramatic signal about a firm's viability. Are CLO managers with bank arranger affiliations also sensitive to these lesser credit risk exposures? These credit downgrades are not followed by default so this subset of borrowers is entirely separate from the default sample. Importantly, CLO managers are not expected to mark to market credit downgrades; thus, these changes in investment ratio stem solely from trading.

We modified the regression model defined by equation (1) so that it will estimate the differential effect of credit downgrades and affiliation on trading by CLO managers. The regression specification is very similar to the index $h = -12, \dots, 0, \dots, +12$, which traces the horizon around a credit downgrading. The sample design treats all multiple firm downgrades as separate events but also controls for firm fixed effects. Figure 3 provides a graphical overview of the regression estimates of the three categories of CLO managers' relationships that we considered before. Consistent with the default findings, the investment ratio of CLO managers affiliated with the arranger of the loan decreases from about 140 percent at 12 months before the downgrade to around 125 percent at 12 months after the event. In contrast to the firm defaults, however, the trading response is not monotonic, as CLO managers initially increase their holdings and then typically sell off about 6 months before the credit downgrade. This more tepid response is not surprising considering that these credit-related actions are less serious than default.

Overall, the regression estimates are similar to the default regression sample (Table 8). CLO managers continue to be sensitive to borrower creditworthiness, avoiding loans rated below CCC+ and purchasing assets mostly in the [BB - B] rating bracket. The formal hypotheses tests (Table 8, bottom panel) generally echo the graphical evidence (Figure 3). CLO managers

affiliated with the arranger of the loan decrease their average position over the entire downgrade episode by about 20 percent; in contrast, for managers that have weaker forms of relationships with the downgraded loan, most of the decline occurs after the downgrade. An early increase in the holdings of bank affiliated managers when the likelihood of rating downgrade was less apparent partly accounts for the muted and insignificant rise before the downgrade (Figure 3). Generally, bank-affiliated CLO managers appear to lower their exposure about 6 months before the downgrade on loans arranged by their affiliated bank.

5. CONCLUSION

The originate-to-distribute model of securitization is an important financial innovation in the capital allocation markets. This paper focuses on collateralized loan obligations (CLOs)—structured-finance vehicles that enable lenders to originate and distribute loans to investors. We use a unique database of CLO trading transactions to investigate the impact that the institutional affiliations of CLO managers' might have on their trading activity. The analysis focuses on the trading behavior of CLO managers affiliated with banks, a connection that gives the managers access to private information held by the banks in their role as loan arrangers in the syndicated market.

Our regression analysis, examining CLO managers' trading of loans to borrowers that have become financially distressed, reveals that the managers affiliated with the bank that arranged those loans sell off their positions before default. Importantly, they continue to divest from these distressed positions after their default. In contrast, CLO managers that do not have a similar affiliation do not divest these credit exposures.

Although these results are consistent with the premise that bank-affiliated CLO managers have access to private information, we cannot exclude the possibility that they are driven by a concern on the part of the managers that CLO positions weakened by distressed loans arranged

by the affiliated bank will damage that bank's reputation. Indeed, on close inspection, we find that such concerns are a significant factor in the decision to divest defaulted loans. Although the value of private information is moot and fully priced after default, we find that bank-affiliated CLO managers are significantly more averse to holding their own arranged problem loans than distressed loans arranged by unaffiliated institutions.

Our findings also show that the nature of the CLO affiliation influences the CLO manager's risk appetite. Even after the default date, bank-affiliated CLO managers divest distressed loans that had been arranged by the parent bank, indicating that concern for the reputation of the bank plays an important role in the trading activity of those managers. Consistent with this insight, a simple analysis of non-rated (NR) residual tranches reveals that bank-affiliated CLOs maintain on average larger equity cushions to absorb investor losses.

Finally, over the last two decades banks have searched for ways to distribute the loans they originate (Bord and Santos 2012). Our findings suggest that reputational concerns pose a potential limit to banks' ability to originate loans and distribute them via their affiliated CLOs.

REFERENCES

- Benmelech, E., J. Dlugosz, and V. Ivashina. (2012) “Securitization without Adverse Selection: The Case of CLOs.” *Journal of Financial Economics* 106, 91–113.
- Bord, V. and J. Santos (2012). “The Rise of the Originate-to-Distribute Model and the Role of Banks in Financial Intermediation.” *Federal Reserve Bank of New York Economic Policy Review* 18, 21–34.
- Bord, V. and J. Santos (2015). “Does Securitization of Corporate Loans Lead to Riskier Lending?” *Journal of Money, Credit and Banking* 47, 415-444.
- Brown, J. (2007). “Understanding Reputational Risk: Identify, Measure, and Mitigate the Risk,” *SRC Insights*. https://www.philadelphiafed.org/bank-resources/publications/src-insights/2007/fourth-quarter/q4si1_07
- Bushman, R., A. Smith, and R. Wittenberg-Moerman (2011). “Price Discovery and Dissemination of Private Information by Loan Syndicate Participants,” *Journal of Accounting Research* (2011), 921-972.
- Cameron C. and D. Miller. “A Practitioner's Guide to Cluster-Robust Inference”, *Journal of Human Resources* 50 (2015), 317-373.
- Carter, R., and S. Manaster (1990). “Initial Public Offerings and Underwriter Reputation” *The Journal of Finance* 45 (1990), 1045-1067
- Carter, R., F. Dark, and A. Singh (1998). “Underwriter Reputation, Initial Returns and Long Run Performance of IPO Stocks,” *The Journal of Finance* 53, 285-311.
- Cetorelli, N., and S. Peristiani (2012). “The Role of Banks in Asset Securitization,” *Federal Reserve Bank of New York Economic Policy Review* 18, 47-63.
- Gopalan, R, V. Nanda, and V. Yerramilli (2011). “Lead Arranger Reputation and the Loan Syndication Market.” Working Paper.
https://www.researchgate.net/publication/251243779_Lead_Arranger_Reputation_and_the_Loan_Syndication_Market
- Guo, Z., and S. Zhang (2015). “The Syndicate Structure of Securitized Corporate Loans,” Working Paper. http://swfa2015.uno.edu/E_Debt_Management/SWFA2015_Paper_146.pdf
- Lou, Y., M. Loumioti, and F. Vasvari (2014). “CLOs’ Trading Activity and Performance,” Working Paper, 2014.
<https://www.kellogg.northwestern.edu/departments/accounting/~media/A41AC10049B4473293086BDA5231775C.ashx>
- Loumioti, M., and F. Vasvari (2016). “Portfolio Performance Manipulation in Collateralized Loan Obligations,” Working Paper.
http://papers.ssrn.com/sol3/papers.cfm?abstract_id=2803704
- Massoud, N., D. Nandy, A. Saunders, and K. Song (2009). “Do Hedge Funds Trade on Private Information? Evidence from Syndicated Lending and Short-Selling,” *Journal of Financial Economics* 99, 477-499.
- Nadauld, T., and M. Weisbach (2014). “Did Securitization Affect the Cost of Corporate Debt?” *Journal of Financial Economics* 105, 332-352.

- Paligorova, T., and J. Santos (2016). “Non-Bank Investors and Loan Renegotiations,” Federal Reserve Bank of New York, Working Paper.
- Perry, J., and P. de Fontnouvelle (2015). “Measuring Reputational Risk: The Market Reaction to Operational Loss Announcements” Working Paper.
<https://www.bostonfed.org/-/media/Documents/ar/PDF/jppd1005.pdf?la=en>
- Peristiani, S., and J. Santos (2015). “Investigating the Trading Activities of CLO Portfolio Managers,” Liberty Street Economics, August 3, 2015.
<http://libertystreeteconomics.newyorkfed.org/2015/08/investigating-the-trading-activity-of-clo-portfolio-managers.html#>.
- Petersen, M. (2004) .“Information: Hard and Soft,” Kellogg School of Management Northwestern University, Working Paper.
<http://www.kellogg.northwestern.edu/faculty/petersen/htm/papers/softhard.pdf>
- Shivdasani, A., and Y. Wang (2011). “Did Structured Credit Fuel the LBO Boom?” *Journal Finance* 66, 1291–1328

Table 1. Collateral manager distribution of the CLO sample, 2007-2011

	Number of CLOs	% of CLOs	Median CLO Balance (\$ Millions)	Number of Institutions	Default Rate
<u>Collateral Manager Type</u>					
Banks	55	23.01	500	22	0.19
Hedge Funds	31	12.97	487	18	0.26
Insurance Firms	19	7.95	423	5	0.29
Investment Firms	98	41.00	494	43	0.22
PE Firms	<u>36</u>	15.06	484	<u>12</u>	0.28
	239			100	
<u>CLO Underwriter Type</u>					
Banks	239				
<u>Number of Borrowers</u>					
Panel Size	5,039				
Borrowers × CLOs	1,471,565				
	65,129				

Notes: The table presents basic summary statistics for the entire cross-sectional time series sample of CLOs. Default rate = the fraction of borrowers that experienced default during the sample period. Number of institutions = unique number firms that manage the CLOs. Number of borrowers = firms with loans held in all CLO portfolios.

Table 2. Cross-tabulation of CLO manager relationship role by institution type for sample of defaulted borrowers

		Banks	Hedge Funds	Insurance Firms	Asset Managers	PE Firms	Total %Column
Manager is arranger	Number	201	3	0	9	2	215
	%Row	93.49	1.4	0	4.19	0.93	
	%Column	9.28	0.19	0	0.18	0.12	1.93
Manager is participant or underwriter is participant	Number	265	169	98	588	133	1,253
	%Row	21.15	13.49	7.82	46.93	10.61	
	%Column	12.23	10.56	12.25	12	8.12	18.08
Underwriter is arranger	Number	395	346	124	896	247	2,008
	%Row	19.67	17.23	6.18	44.62	12.3	
	%Column	18.24	21.63	15.5	18.29	15.08	11.28
No relationship	Number	1305	1082	578	3,407	1,256	7,628
	%Row	17.11	14.18	7.58	44.66	16.47	
	%Column	60.25	67.63	72.25	69.53	76.68	68.71
Total	Total	2,166	1,600	800	4,900	1,638	11,104
	%Row	19.51	14.41	7.2	44.13	14.75	100

Notes. The table cross-tabulates the frequency of *defaulted borrower* relationships by institutional type of CLO manager.

Table 3. Summary Statistics for Panel Sample of Defaulted Borrowers, 2007-2011

REGRESSION VARIABLES (Panel index dimensions: month (t), CLO (j), borrower (i))	Mean	Median	1 st Pctl	99 th Pctl
DEPENDENT VARIABLE				
Investment Ratio R_{tij} = Aggregate position held at month (t) divided by the initial position (t=0) (percent)	136.9	99.24	0.21	10,000
EXPLANATORY VARIABLES				
<u>Borrower/Loan Characteristics</u> x_{tij}^B				
POSITION = Nominal position held (\$ millions)	3.81	2.41	0.002	25.75
US_FIRM = 1 if borrower is US domiciled	0.95			
%ISSUE_OWNED = Percent of entire issue owned by CLO	4.25	0.25	0.0022	37.9
MATURITY = Weighted remaining loan maturity (years)	4.25	4.00	0.050	7.00
EQUITY_REL = 1 if manager has equity relationships with borrower	0.04			
<i>Rating Indicators</i>				
AAA to BBB-	0.0003			
BB+	0.0024			
BB	0.0151			
BB-	0.0163			
B+	0.0580			
B	0.2029			
B-	0.2177			
CCC+	0.2484			
CCC	0.1108			
CCC-	0.0718			
CC+ to C	0.0563			
<i>Manager-Borrower Loan Relationships</i>				
No Relationship, $NORel_{tji}$	0.671			
Manager is loan arranger, Rel_{tji}^M	0.029			
Underwriter is Arranger/Participant, Rel_{tji}^U	0.310			
• Underwriter is loan arranger	0.198			
• Underwriter or manager are loan participants	0.112			
<u>CLO Characteristics</u> x_{tij}^{CLO}				
CLO_BALANCE = CLO balance (\$ millions)	573	487	258.5	2,018
Number of Defaulted Borrowers in Panel	376			
Number of Observations in Panel	148,486			

Notes: This table summarizes the explanatory and dependent variable of the regression sample. All continuous explanatory variables are winsorized at the 1- and 99-percentile level. The regression samples are smaller because they are restricted to the pertinent subsets of borrowers. The overall sample size indicates the maximum number observations available for most regression variables.

Table 4. Influence of loan relationships on investment ratio 12-months before and after default

$$R_{tj} = \alpha_t + \alpha_i + \beta_1 x_{tj}^B + \beta_2 x_{tj}^{CLO} + \sum_{h=-12}^{12} \delta_h^M \text{Rel}_{tj,h}^M + \sum_{h=-12}^{12} \delta_h^U \text{Rel}_{tj,h}^U + \sum_{h=-12}^{12} \delta_h^N \text{NOrel}_{tj,h} + \varepsilon_{tj}$$

Dependent variable: Borrower investment ratio R_{tj} (percent)

	All Borrowers	Manager- Arranger Borrowers	Single Security Loans
Log(POSITION)	69.25*** (117.97)	84.40*** (78.49)	53.35*** (80.72)
%ISSUE_OWNED	-2.32*** (-25.54)	-1.97*** (-7.32)	-2.17*** (-22.46)
MATURITY	-2.23*** (-6.50)	-8.87*** (-14.20)	0.50 (1.08)
US_FIRM	-4.67* (-1.85)	-1.19 (-0.22)	-6.93** (-1.97)
CLO_BALANCE	-0.03*** (-36.80)	-0.04*** (-23.63)	-0.02*** (-21.97)
EQUITY_REL	14.62*** (9.55)	16.10*** (5.70)	10.45*** (6.21)
AAA to BBB-	20.13*** (3.14)	387.87*** (15.08)	107.36*** (8.09)
BB+	4.06 (0.91)	18.77*** (2.87)	-22.03*** (-4.42)
BB	-8.48*** (-2.62)	-9.56* (-1.95)	-8.85** (-2.26)
BB-	19.62*** (10.24)	38.31*** (11.50)	9.89*** (4.81)
B+	1.46 (1.12)	5.77** (2.30)	2.35 (1.57)
B	1.71 (1.53)	2.82 (1.27)	1.81 (1.54)
B-	-1.23 (-1.19)	-1.53 (-0.70)	0.61 (0.55)
CCC+	0.42 (0.35)	0.50 (0.19)	2.19* (1.76)
CCC	-1.62 (-1.20)	-1.08 (-0.35)	2.18* (1.65)
CCC-	-9.80*** (-6.20)	-6.18 (-1.47)	-5.18*** (-3.18)
CC to C	-13.64*** (-5.51)	-12.83** (-2.25)	-3.00 (-1.14)

Table continues next page

<i>Hypothesis 1A. Are managers selling before default?</i>		$\bar{\delta}_B^I$	
Manager-arranger relationships $\bar{\delta}_B^M \leq 0$	-11.70***	-8.58**	-19.26***
Managers with underwriter/participant relationships $\bar{\delta}_B^U \leq 0$	-1.56	-2.50	-2.68**
Managers with no relationships $\bar{\delta}_B^N \leq 0$	2.62***	8.23***	2.05***
<i>Hypothesis 1B. Who is more responsive before default?</i>		$\bar{\delta}_B^I - \bar{\delta}_B^K$	
Manager-arranger is more responsive than underwriter/participant $\bar{\delta}_B^M - \bar{\delta}_B^U \leq 0$	-10.15***	-6.08	-16.58***
Manager-arranger is more responsive than no-relationship manager $\bar{\delta}_B^M - \bar{\delta}_B^N \leq 0$	-14.32***	-16.81***	-21.31***
<i>Hypothesis 2A. Are managers selling after default?</i>		$\bar{\delta}_A^I$	
Manager-arranger with relationships $\bar{\delta}_A^M \leq 0$	-19.69***	-17.75***	-24.06***
Managers with underwriter/participant relationships $\bar{\delta}_A^U \leq 0$	2.81**	3.20*	3.82***
Managers with no relationships $\bar{\delta}_A^N \leq 0$	1.88**	5.06***	0.82
<i>Hypothesis 2B. Who is more responsive after default?</i>		$\bar{\delta}_A^I - \bar{\delta}_A^K$	
Manager-arranger is more responsive underwriter/participant $\bar{\delta}_A^M - \bar{\delta}_A^U \leq 0$	-22.50***	-20.96***	-27.88***
Manager-arranger is more responsive than no-relationship manager $\bar{\delta}_A^M - \bar{\delta}_A^N \leq 0$	-21.58***	-22.82***	-24.87***
Borrower Fixed Effects	Yes	Yes	Yes
Year and Month Time Effects	Yes	Yes	Yes
Number Observations	141,124	67,845	77,322
Number of Borrowers	374	67	353
R ²	0.23	0.16	0.23

Notes: The table presents the regression estimates of the investment ratio regression model formally define by equation (1). The regression sample is limited only to borrowers that had at least 6-month history before and after default. The first column summarizes the estimate of the entire CLO sample. The second column focuses only on borrowers that have an extant loan relationship with CLO managers. The last column examines only single loan positions. The hypothesis tests panel presents the average value of the maintained hypothesis with its corresponding F-value significance. The null hypotheses of $\bar{\delta}_B^I \leq 0$ or $\bar{\delta}_B^I - \bar{\delta}_B^K \leq 0$ are accepted when the values are negative and statistically significant. When the values are positive and statistically significant the null is rejected in favor of the alternative. The regression panel model was estimated using a GLS feasible approach that corrects for borrower clustered error heteroskedasticity (Cameron and Miller 2015). The symbols (*), (**), and (***) indicated statistical significance at the 10-, 5- and 1-percent level.

Table 5. Influence of loan relationships on investment ratio 6-months before and after default

$$R_{tj} = \alpha_t + \alpha_i + \beta_1 x_{tj}^B + \beta_2 x_{tj}^{CLO} + \sum_{h=-6}^6 \delta_h^M \text{Rel}_{tj,h}^M + \sum_{h=-6}^6 \delta_h^U \text{Rel}_{tj,h}^U + \sum_{h=-6}^6 \delta_h^N \text{NORel}_{tj,h} + \varepsilon_{tj}$$

Dependent variable: Borrower investment ratio R_{tj} (percent)

	All Borrowers	Manager- Arranger Borrowers	Single Security Loans
Log(POSITION)	69.26*** (118.14)	84.37*** (78.58)	53.30*** (80.71)
%ISSUE_OWNED	-2.31*** (-25.47)	-1.95*** (-7.26)	-2.16*** (-22.59)
MATURITY	-2.23*** (-6.53)	-8.88*** (-14.24)	0.47 (1.02)
US_FIRM	-4.52* (-1.79)	-0.86 (-0.16)	-6.82* (-1.93)
CLO_BALANCE	-0.03*** (-36.76)	-0.04*** (-23.52)	-0.02*** (-21.83)
EQUITY_REL	14.64*** (9.56)	16.22*** (5.74)	10.68*** (6.34)
AAA to BBB-	20.40*** (3.20)	385.71*** (15.02)	107.18*** (8.08)
BB+	3.99 (0.90)	17.17*** (2.65)	-21.89*** (-4.42)
BB	-8.34*** (-2.59)	-9.58* (-1.96)	-8.74** (-2.24)
BB-	19.86*** (10.38)	38.39*** (11.58)	9.92*** (4.84)
B+	1.62 (1.24)	5.82** (2.35)	2.54* (1.70)
B	1.81 (1.63)	3.03 (1.37)	1.93* (1.65)
B-	-1.19 (-1.16)	-1.88 (-0.88)	0.69 (0.62)
CCC+	0.56 (0.47)	1.08 (0.41)	2.31* (1.86)
CCC	-1.45 (-1.08)	-0.49 (-0.16)	2.23 * (1.69)
CCC-	-9.84*** (-6.22)	-6.04 (-1.46)	-5.18*** (-3.20)
CC to C	-13.51*** (-5.47)	-12.72** (-2.23)	-2.85 (-1.08)

Table continues next page

<i>Hypothesis 1A. Are managers selling before default?</i>		$\bar{\delta}_B^l$	
Manager-arranger relationships $\bar{\delta}_B^M \leq 0$	-14.68***	-11.60**	-16.72***
Manager underwriter/participant relationships $\bar{\delta}_B^U \leq 0$	-0.72	-0.36	-2.31
Managers with no relationships $\bar{\delta}_B^N \leq 0$	1.51**	5.23***	0.95
<i>Hypothesis 1B. Who is more responsive before default?</i>		$\bar{\delta}_B^l - \bar{\delta}_B^k$	
Manager-arranger vs underwriter/participant $\bar{\delta}_B^M - \bar{\delta}_B^U \leq 0$	-13.96***	-11.24**	-14.40***
Manager-arranger vs underwriter $\bar{\delta}_B^M - \bar{\delta}_B^N \leq 0$	-16.19***	-16.83***	-17.67***
<i>Hypothesis 2A. Are managers selling after default?</i>		$\bar{\delta}_A^l$	
Managers with no relationships $\bar{\delta}_A^M \leq 0$	-13.38***	-11.41**	-20.74***
Manager-arranger relationships $\bar{\delta}_A^U \leq 0$	2.47*	1.82	3.92***
Manager-underwriter arranger $\bar{\delta}_A^N \leq 0$	1.74**	4.49**	0.41
<i>Hypothesis 2A. Who is more responsive after default?</i>		$\bar{\delta}_A^l - \bar{\delta}_A^k$	
Manager-arranger vs underwriter $\bar{\delta}_A^M - \bar{\delta}_A^U \leq 0$	-15.85 ***	-13.23**	-24.66***
Manager-arranger vs underwriter $\bar{\delta}_A^M - \bar{\delta}_A^N \leq 0$	-15.12 ***	-15.91***	-21.15***
Borrower Fixed Effects	Yes	Yes	Yes
Year and Month Time Effects	Yes	Yes	Yes
Number Observations	141,124	67,845	77,322
Number of Borrowers	374	66	353
R ²	0.23	0.16	0.23

Notes: The table presents the regression estimates of the investment ratio regression model formally define by equation (1). The regression sample is limited only to borrowers that had at least 6-month history before and after default. The first column summarizes the estimate of the entire CLO sample. The second column focuses only on borrowers that have an extant loan relationship with CLO manager. The last column examines only single loan positions. The regression panel model was estimated using a GLS feasible approach that accounts for borrower clustered error heteroskedasticity (Cameron and Miller 2015). The hypothesis tests panel presents the average value of the maintained hypothesis with its corresponding F-value significance. The null hypotheses of $\bar{\delta}_B^l \leq 0$ or $\bar{\delta}_B^l - \bar{\delta}_B^k \leq 0$ are accepted when the values are negative and statistically significant. When the values are positive and statistically significant the null is rejected in favor of the alternative. The symbols (*), (**), and (***) indicated statistical significance at the 10-, 5-, and 1-percent level.

Table 6. Importance of regulatory structure: Response of bank and nonbank managers

$$R_{ij} = \alpha_i + \alpha_t + \beta_1 x_{ij}^B + \beta_2 x_{ij}^{CLO} + \sum_{h=-12}^{12} \delta_h^{BANK} Rel_{ij,h}^{BANK} + \sum_{h=-12}^{12} \delta_h^{NONBANK} Rel_{ij,h}^{NONBANK} + \varepsilon_{ij}.$$

Dependent variable: Borrower investment ratio R_{ij} (percent)

	All Borrowers	Borrowers Without Relationships	Borrowers With Relationships
Log(POSITION)	69.30*** (118.04)	65.75*** (102.08)	80.76*** (75.03)
%ISSUE_OWNED	-2.32*** (-25.67)	-2.15*** (-22.70)	-1.86*** (-7.63)
MATURITY	-2.17*** (-6.36)	-0.92** (-2.48)	-7.01*** (-10.35)
US_FIRM	-4.58* (-1.81)	-2.10 (-0.79)	-11.73** (-2.10)
CLO_BALANCE	-0.03*** (-37.00)	-0.03*** (-31.77)	-0.04*** (-21.61)
EQUITY_REL	14.34*** (9.35)	16.04*** (9.51)	7.31** (2.41)
AAA to BBB-	20.00*** (3.14)	17.93*** (2.87)	-33.25 (-1.05)
BB+	3.84 (0.87)	-11.41** (-1.98)	30.72*** (4.78)
BB	-8.31*** (-2.59)	-4.36 (-1.08)	-9.02* (-1.83)
BB-	19.88*** (10.39)	10.37*** (4.77)	32.15*** (10.12)
B+	1.64 (1.26)	0.95 (0.68)	3.79* (1.66)
B	1.69 (1.51)	-0.34 (-0.29)	6.24*** (3.11)
B-	-1.22 (-1.19)	-1.36 (-1.29)	-1.81 (-0.94)
CCC+	0.60 (0.50)	-1.25 (-1.01)	1.49 (0.67)
CCC	-1.32 (-0.99)	-1.13 (-0.82)	-1.88 (-0.72)
CCC-	-9.77*** (-6.17)	-9.62*** (-6.04)	6.85** (2.09)
CC to C	-13.12*** (-5.33)	-14.17*** (-5.09)	-8.81** (-2.26)

Table continues next page

<i>Hypothesis 1A. Are managers selling before default?</i>		$\bar{\delta}_B^1$	
Bank Managers $\bar{\delta}_B^{\text{BANK}} \leq 0$	-5.11***	-1.92	-7.13***
Nonbank Managers $\bar{\delta}_B^{\text{NONBANK}} \leq 0$	1.00	0.94	0.82
<i>Hypothesis 1B. Who is more responsive before default?</i>		$\bar{\delta}_B^{\text{BANK}} - \bar{\delta}_B^{\text{NONBANK}}$	
Bank Managers vs Nonbank Managers $\bar{\delta}_B^{\text{BANK}} - \bar{\delta}_B^{\text{NONBANK}} \leq 0$	-6.11***	-2.86**	-7.95***
<i>Hypothesis 2A. Are managers selling after default?</i>		$\bar{\delta}_A^1$	
Bank Managers $\bar{\delta}_A^{\text{BANK}} \leq 0$	-4.74***	-2.66**	-5.45**
Nonbank Managers $\bar{\delta}_A^{\text{NONBANK}} \leq 0$	0.69	0.42	1.66
<i>Hypothesis 2B. Who is more responsive after default?</i>		$\bar{\delta}_A^{\text{BANK}} - \bar{\delta}_A^{\text{NONBANK}}$	
Bank Managers vs Nonbank Managers $\bar{\delta}_A^{\text{BANK}} - \bar{\delta}_A^{\text{NONBANK}} \leq 0$	-5.43***	-3.07**	-7.11***
Borrower Fixed Effects	Yes	Yes	Yes
Year and Month Time Effects	Yes	Yes	Yes
Number Observations	141,124	94,560	46,564
Number of Borrowers	374	358	156
R^2	0.23	0.26	0.26

Notes: The table summarizes the regression estimates of the investment ratio specification defined by equation (3). The sample is limited only to borrowers that had at least 6-month history before and after default. The first column presents the estimates over the entire CLO sample. The third and second columns focus on borrowers with and without relationships with CLO manager or underwriters, respectively. The regression panel was estimated using a GLS feasible approach that accounts for borrower clustered error heteroskedasticity (Cameron and Miller 2015). The hypothesis tests panel presents the average value of the maintained hypothesis with its corresponding F-value significance. The null hypotheses of $\bar{\delta}_\bullet^1 \leq 0$ or $\bar{\delta}_\bullet^1 - \bar{\delta}_\bullet^2 \leq 0$ are accepted when the values are negative and statistically significant. When the values are positive and statistically significant the null is rejected in favor of the alternative. The symbols (*), (**), and (***) indicated statistical significance at the 10-, 5-, and 1-percent level.

Table 7. The importance of reputational risk: Evidence from borrowers without any loan relationships

$$R_{ij} = \alpha_i + \alpha_t + \beta_1 x_{ij}^B + \beta_2 x_{ij}^{CLO} + \sum_{h=-12}^{12} \delta_h^M \text{Rel}_{ij,h}^M + \sum_{h=-12}^{12} \delta_h^U \text{Rel}_{ij,h}^U + \sum_{h=-12}^{12} \delta_h^{N-BANK} \text{NOrel}_{ij,h}^{BANK} + \sum_{h=-12}^{12} \delta_h^{N-NONBANK} \text{NOrel}_{ij,h}^{NONBANK} + \varepsilon_{ij}$$

Dependent variable: Borrower investment ratio R_{ij} (percent)

Explanatory Variables	All Firms
Log(POSITION)	69.41*** (118.01)
%ISSUE_OWNED	-2.34*** (-25.93)
MATURITY	-2.18*** (-6.33)
US_FIRM	-4.75* (-1.87)
CLO_BALANCE	-0.03*** (-36.99)
EQUITY_REL	13.14*** (8.53)
AAA to BBB-	18.88*** (2.93)
BB+	3.66 (0.82)
BB	-8.97*** (-2.78)
BB-	19.20*** (10.00)
B+	1.41 (1.07)
B	1.51 (1.35)
B-	-1.24 (-1.21)
CCC+	0.32 (0.27)
CCC	-1.71 (-1.27)
CCC-	-9.65*** (-6.10)
CC to C	-13.65*** (-5.52)

<i>Hypothesis 1A. Are managers selling before default?</i>	$\bar{\delta}_B^i$
Manager-arranger relationships $\bar{\delta}_B^M \leq 0$	-12.07***
Managers with underwriter/participant relationships $\bar{\delta}_B^U \leq 0$	-1.63
Bank managers without relationships $\bar{\delta}_B^{N-BANKS} \leq 0$	-2.11
Nonbank managers without relationships $\bar{\delta}_B^{N-NONBANKS} \leq 0$	3.46
<i>Hypothesis 1B. Who is more responsive before default?</i>	$\bar{\delta}_B^i - \bar{\delta}_B^k$
Manager-arranger vs banks without relationships $\bar{\delta}_B^M - \bar{\delta}_B^{N-BANK} \leq 0$	-9.95***
Manager-arranger vs nonbanks without relationships $\bar{\delta}_B^M - \bar{\delta}_B^{N-NONBANK} \leq 0$	-15.53***
Without relationships: bank vs nonbank $\bar{\delta}_B^{N-BANK} - \bar{\delta}_B^{N-NONBANK} \leq 0$	-5.57***
<i>Hypothesis 2A. Are managers selling after default?</i>	$\bar{\delta}_A^i$
Manager-arranger relationships $\bar{\delta}_A^M \leq 0$	-19.96***
Managers with underwriter/participant relationships $\bar{\delta}_A^U \leq 0$	2.88**
Bank managers without relationships $\bar{\delta}_A^{N-BANKS} \leq 0$	-3.10
Nonbank managers without relationships $\bar{\delta}_A^{N-NONBANKS} \leq 0$	2.61
<i>Hypothesis 2B. Who is more responsive after default?</i>	$\bar{\delta}_A^i - \bar{\delta}_A^k$
Manager-arranger vs underwriter/participant managers $\bar{\delta}_A^M - \bar{\delta}_A^U \leq 0$	-22.84***
Manager-arranger vs banks without relationships $\bar{\delta}_A^M - \bar{\delta}_A^{N-BANK} \leq 0$	-16.87***
Manager-arranger vs nonbanks without relationships $\bar{\delta}_A^M - \bar{\delta}_A^{N-NONBANK} \leq 0$	-22.57***
Without relationships: bank vs nonbank $\bar{\delta}_A^{N-BANK} - \bar{\delta}_A^{N-NONBANK} \leq 0$	-5.70***
Borrower Fixed Effects	Yes
Year and Month Time Effects	Yes
Number Observations	141,124
Number of Borrowers	374
R ²	0.23

Notes: The regression presents the regression estimates of the investment ratio regression model formally define by equation (2). The specification focuses on the borrowers without existing loan relationships with CLO managers and underwriters. The sample is limited only to borrowers that had at least 6-month history before and after default. The regression panel model was estimated using a GLS feasible approach that accounts for borrower clustered error heteroskedasticity (Cameron and Miller 2015). The hypothesis tests panel presents the average value of the maintained hypothesis with its corresponding F-value significance. The null hypotheses of $\bar{\delta}_i^i \leq 0$ or $\bar{\delta}_i^i - \bar{\delta}_i^k \leq 0$ are accepted when the values are negative and statistically significant. When the values are positive and statistically significant the null is rejected in favor of the alternative. The symbols (*), (**), and (***) indicated statistical significance at the 10-, 5-, and 1-percent level.

Table 8. Influence of loan relationships for downgraded borrowers (excluding defaults)

$$R_{ij} = \alpha_t + \alpha_i + \beta_1 x_{ij}^B + \beta_2 x_{ij}^{CLO} + \sum_{h=-12}^{12} \delta_h^M \text{Rel}_{ij,h}^M + \sum_{h=-12}^{12} \delta_h^U \text{Rel}_{ij,h}^U + \sum_{h=-12}^{12} \delta_h^N \text{NORel}_{ij,h} + \varepsilon_{ij}.$$

Dependent variable: Borrower investment ratio R_{ij} (percent)

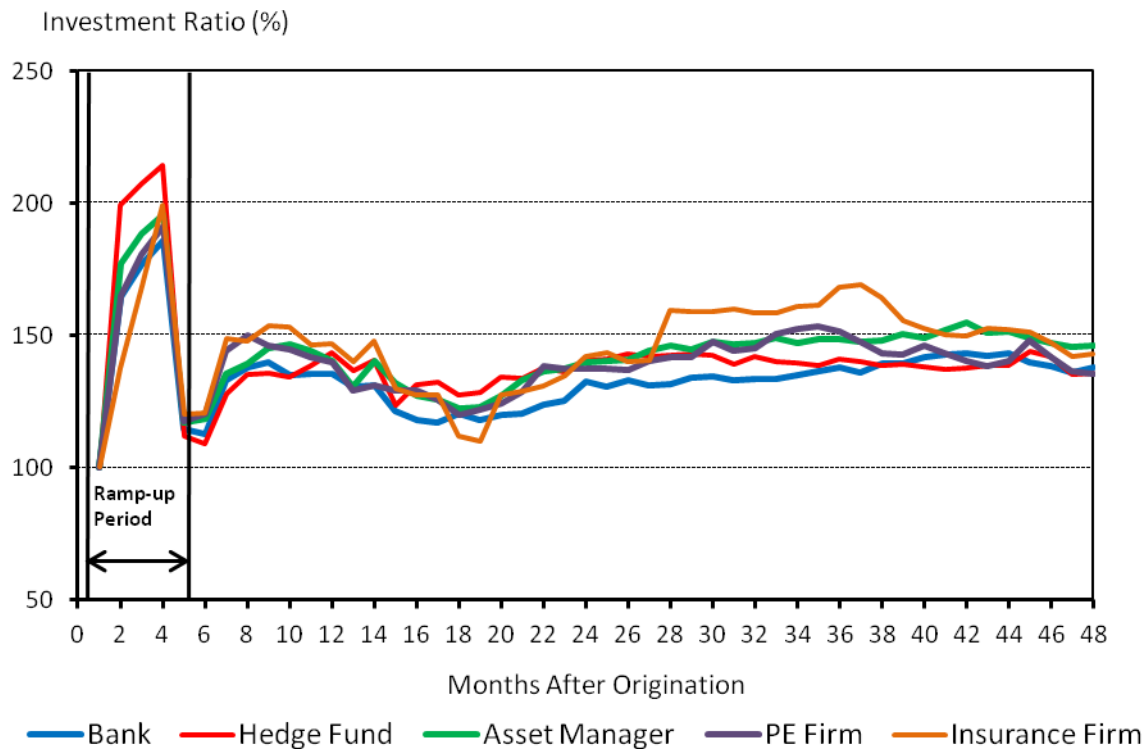
Explanatory Variables	12-Month Horizon
Log(Position)	85.50*** (108.44)
%Issue Owned	-2.95*** (-10.88)
Loan Maturity	-2.02*** (-4.84)
US Firm	12.66** (2.01)
CLO Balance	-0.04*** (-30.33)
EQUITY_REL	-15.14*** (-6.64)
AAA to BBB-	0.85 (0.15)
BB+	-0.76 (-0.19)
BB	7.02* (1.96)
BB-	24.01*** (7.78)
B+	14.55*** (5.03)
B	10.33*** (3.68)
B-	3.03 (1.02)
CCC+	-4.46 (-1.28)
CCC	-18.94*** (-4.81)
CCC-	-26.86*** (-6.53)
CC to C	-18.65*** (-2.79)

Table continues below

<i>Hypothesis 1A: Are managers selling before downgrade?</i>	$\bar{\delta}_B^i$
Manager-arranger relationships $\bar{\delta}_B^M \leq 0$	-1.91
Managers with no relationships $\bar{\delta}_B^N \leq 0$	1.07
Managers with underwriter/participant relationships $\bar{\delta}_B^U \leq 0$	3.02**
<i>Hypothesis 1B. Who is more responsive before downgrade?</i>	$\bar{\delta}_B^i - \bar{\delta}_B^k$
Manager-arranger vs Underwriter/participant $\bar{\delta}_B^M - \bar{\delta}_B^U \leq 0$	-4.93
Manager-arranger vs No relationships $\bar{\delta}_B^M - \bar{\delta}_B^N \leq 0$	-2.98
<i>Hypothesis 2A. Are managers selling after downgrade?</i>	$\bar{\delta}_A^i$
Manager-arranger relationships $\bar{\delta}_A^M \leq 0$	-14.86***
Managers with no relationships $\bar{\delta}_A^N \leq 0$	5.86***
Managers with underwriter/participant relationships $\bar{\delta}_A^U \leq 0$	16.98***
<i>Hypothesis 2B. Who is more responsive after downgrade?</i>	$\bar{\delta}_A^i - \bar{\delta}_A^k$
Manager-arranger vs Underwriter/participant $\bar{\delta}_A^M - \bar{\delta}_A^U \leq 0$	-31.84***
Manager-arranger vs No relationships $\bar{\delta}_A^M - \bar{\delta}_A^N \leq 0$	-20.72***
Borrower Fixed Effects	Yes
Year and Month Time Effects	Yes
Number Observations	185,224
Borrowers	846
R ²	0.20

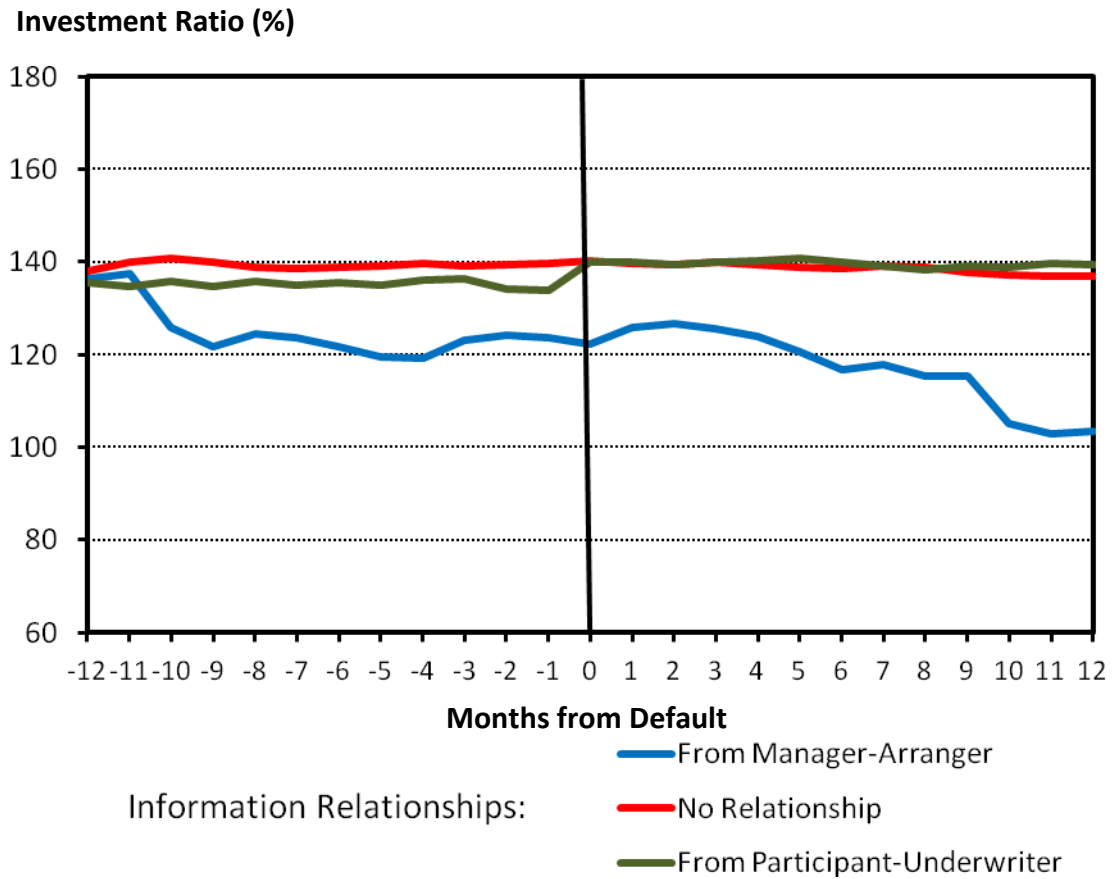
Notes: The table presents the regression estimates of the investment ratio specification defined by equation (1) where the distress event is a credit downgrade. All defaulted borrowers were excluded from the sample. The clusters for downgraded firms are sometimes very small. To avoid small-sample biases, the coefficients were estimated using OLS instead of the feasible GLS estimator applied previously. For the same reason, we also do not use robust standard errors correction; overall, the statistical significance of the coefficients with the robust error correction and the standard OLS errors was very similar. The hypothesis tests panel presents the average value of the maintained hypothesis with its corresponding F-value significance. The null hypotheses of $\bar{\delta}_\bullet^i \leq 0$ or $\bar{\delta}_\bullet^i - \bar{\delta}_\bullet^k \leq 0$ are accepted when the values are negative and statistically significant. When the values are positive and statistically significant the null is rejected in favor of the alternative. The symbols (*), (**), and (***) indicated statistical significance at the 10-, 5-, and 1-percent level.

Figure 1. CLO average investment ratio R_t by collateral manager type



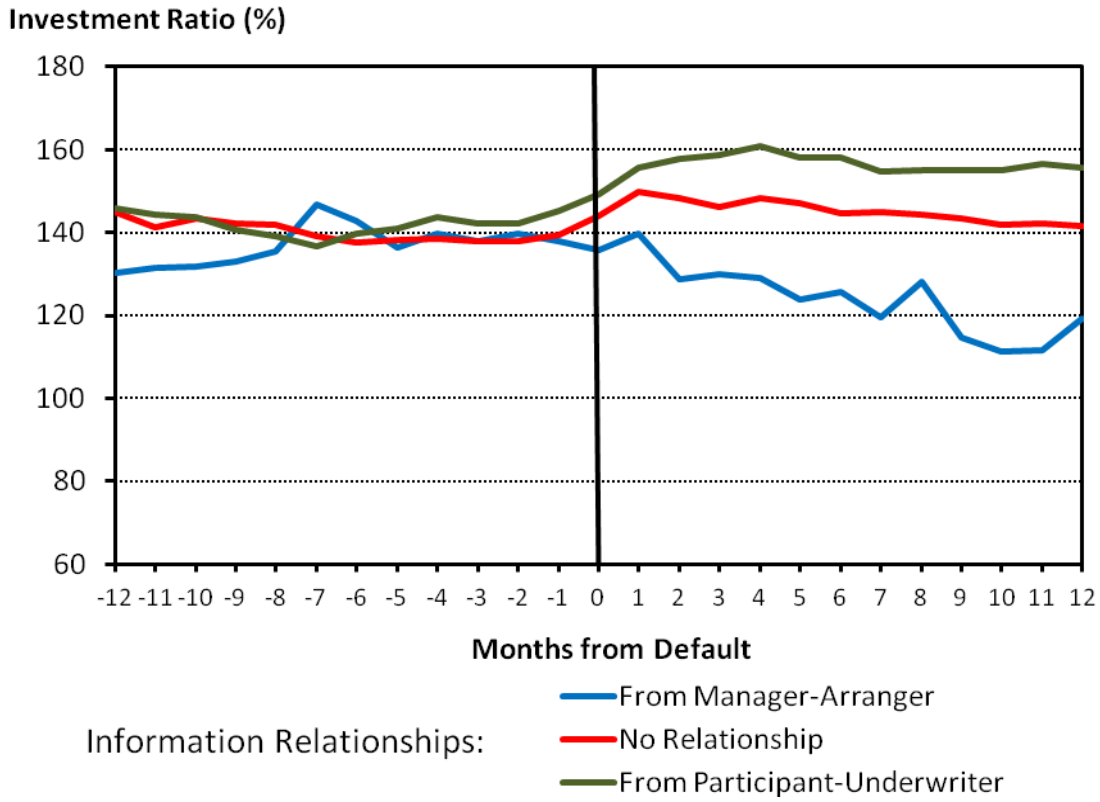
Notes: The figure traces the average investment ratio position held by collateral manager type for the first 4 years of the life of the CLO.

Figure 2. Predicted investment ratio $\hat{R}_h^i$ before and after default by relationship type



Notes: The figure traces the regression predicted investment ratio around default by the manager-underwriter relationship types. The prediction traces the investment ratio for the sample average for each relationship type defined by $\hat{R}_h^i = \bar{\alpha}_i + \bar{\alpha}_t + \hat{\beta}_1 \bar{x}^B + \hat{\beta}_2 \bar{x}^{CLO} + \hat{\delta}_h^i = \hat{\mu} + \hat{\delta}_h^i$ (see equation 1).

Figure 3. Investment ratio prediction $\hat{R}_h^t$ before and after downgrade



Notes: The figure traces the regression predicted investment ratio around downgrade by the manager-underwriter relationship types. The prediction traces the investment ratio for the sample average for each relationship type defined by $\hat{R}_h^t = \bar{\alpha}_t + \bar{\alpha}_t + \hat{\beta}_1 \bar{x}^B + \hat{\beta}_2 \bar{x}^{CLO} + \hat{\delta}_h^t = \hat{\mu} + \hat{\delta}_h^t$ (see equation 1).